

3 (a)	force = rate of change of momentum (allow symbols if defined)	B1	[1]
(b) (i)	$\Delta p = 140 \times 10^{-3} \times (5.5 + 4.0)$ $= 1.33 \text{ kg m s}^{-1}$	C1 A1	[2]
(ii)	force = $1.33 / 0.04$ $= 33.3 \text{ N}$	M1 A0	[1]
(c) (i)	taking moments about B $(33 \times 75) + (0.45 \times g \times 25) = F_A \times 20$ $F_A = 129 \text{ N}$	C1 C1 A1	[3]
(ii)	$F_B = 33 + 129 + 0.45g$ $= 166 \text{ N}$	C1 A1	[2]

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